

Inyoung Oh

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Korea Institute of Science and Technology (KIST), Seoul, Republic of Korea
Fusing semantic and geometric cues for reliable, metric 3D spatial perception.

Research Focus

My research makes learned and semantic 3D *metric, consistent, and structured* by fusing it with explicit geometry, recovering reliable spatial structure exactly where data-driven models are least faithful: at object boundaries, sharp discontinuities, and absolute scale.

- Scene-intrinsic geometric cues (surface normals, sharp boundaries, ground planes) as transferable structure injected into learned and semantic 3D
- Metric, consistent 3D from a single image: bridging 2D and 3D through scene geometry, where learned depth and multi-view reconstruction fall short
- Geometric representations that generalize across sensors and domains (CAD, LiDAR, indoor/outdoor scans, monocular)

Education

Ph.D., Mechanical and Robotics Engineering, Gwangju Institute of Science and Technology (GIST) **Feb 2026**

Dissertation: *Normal Vector Estimation, and Semantic Segmentation of 3D Point Clouds using Deep Learning and Geometric Analysis*

M.S., Mechatronics, GIST **Feb 2018**

Thesis: *Sphere and Cylinder Detection in Kinect Point Clouds using RANSAC and the 2D Hough Transform*

B.S., Mechatronics, Chungnam National University **Feb 2016**

Research Experience

Postdoctoral Fellow, Center for AI, KIST | Advisor: Dr. Junghyun Cho **Feb 2026–Present**

- Reframed metric relocation from a moving camera as a **geometric 2D–3D bridge**: on an up-to-scale map, metric placement reduces to a ray-ground intersection set by one scene-intrinsic quantity, the ground-plane normal, independent of pose and reconstruction method.
- The geometric estimator recovers metric position **7.7–100×** better than monocular-depth and VGGT-style feed-forward baselines, even given ground-truth pose, across indoor and outdoor capture (in preparation [\[P1\]](#)).

Research Assistant, Modeling and Simulation Lab, GIST | Advisor: Prof. Kwang Hee Ko **Feb 2016–Feb 2026**

Established the geometric half of this fusion: explicit scene-intrinsic cues (surface normals, sharp features), injected into learned models, that transfer across sensors, sampling density, and noise.

Sharp Feature Detection & Geometric Descriptors

- **SFD-Net** (ECCV 2026): an architecture-agnostic descriptor detecting sharp features as surface-normal discontinuities; prepended to three backbones *unmodified* it adds +4–7 pp Average F1, transfers *zero-shot* from 10K-point CAD to scans $\sim 36\times$ larger, and is SOTA on ABC across all four noise conditions. [\[C1\]](#)
- Replacing its heuristic scales with a training-free directional-statistics prior that stays informative at discontinuities (in preparation). [\[P2\]](#)

Normal Vector Estimation

- Multi-scale normal estimator with attention-based scale weighting and Chamfer-distance noise-aware supervision; SOTA on the PCPNet primitive subset (noise-free RMSE 1.821°). [\[J1\]](#)

Semantic Segmentation (Outdoor & Indoor)

- Injected intensity-assisted normals into RandLA-Net via a Normal Local Feature Aggregation module: +4.0 pp mIoU on SemanticKITTI (53.9→57.9), with the largest gains on safety-critical classes (e.g. Bicycle +21.4 pp). [\[J3\]](#)
- Extended refined normals into boundary-aware indoor segmentation: +6.3% mIoU over the PushBoundary baseline on S3DIS (Ph.D. thesis).

6-DoF Pose Estimation & MR Systems

- Designed RoI-PCA color augmentation decoupling object appearance from background: +10.77 pp 5cm5° accuracy on LINEMOD; integrated into a real-time, markerless multi-device MR collaboration system. [\[J2\]](#)

Geometric Primitive Recognition

- Prior-free pipeline recognition from raw point clouds (curvature-driven classification + sphere-RANSAC centerline recovery), with no user priors for radius, orientation, or connectivity. [\[J4\]](#)

Publications

Published / Accepted

[\[C1\]](#) Inyoung Oh; K.H. Ko. “SFD-Net: Sharp Feature Detection Network Based on Local Geometric Features.” *European Conference on Computer Vision (ECCV)*, 2026. (accepted) [\[project\]](#) [\[code\]](#)

[J1] Inyoung Oh; J. Song; M. Kim; D. Yun; K.H. Ko. “Attention-Guided Multi-Scale Point Cloud Normal Estimation with Noise-Aware Training.” *Int. J. Image and Graphics*, **2026**. (*accepted*)

[J2] Inyoung Oh; G. Jang; J. Song; M. Son; D. Kim; J. Yun; K.H. Ko. “A Mixed Reality-based Remote Collaboration Framework Using Improved Pose Estimation.” *Computers in Industry*, 174, 104414, **2026**.

[J3] M. Kim; Inyoung Oh; D. Yun; K.H. Ko. “Improved Semantic Segmentation Network Using Normal Vector Guidance for LiDAR Point Clouds.” *J. Computational Design and Engineering*, 10(6), 2332–2344, **2023**.

[J4] Inyoung Oh; K.H. Ko. “Automated Recognition of 3D Pipelines from Point Clouds.” *The Visual Computer*, 37(6), 1385–1400, **2021**.

In Preparation

[P1] Inyoung Oh et al. “Metric Relocalization in Up-to-Scale Maps via Ground-Plane Geometry.”

[P2] Inyoung Oh et al. “A Directional Statistics Approach to Geometric Sharpness Estimation in 3D Point Clouds.”

Patent

K.H. Ko; Inyoung Oh. “Automated System for Detecting 3D Pipeline.” KR 2010-2135828 B1 (Registered), July 2020.

Awards & Scholarships

Best Poster Award (×5), Korean Society for Computational Design and Engineering (CDE)	2020–2025
CDE DX Encouragement Award, Korean CDE Society	2022
Outstanding PhD Student RA Scholarship (×6), GIST	2018–2025
GIST Full Scholarship (Government-funded, Doctoral & Master’s)	2016–2026

Invited Seminars & Presentations

Invited seminar, KEPCO Research Institute: “Object-detection-based 3D LiDAR intensity calibration and real-time detection of clustered object candidates.”	2021
Inyoung Oh; K.H. Ko. “Automatic Detection of Cylindrical Objects from Unorganized Point Cloud.” <i>EuroVR</i> (poster), London, UK.	2018
Inyoung Oh; K.H. Ko. “A RANSAC-based Method for Detection of Multiple Spheres from a Point Cloud.” <i>Computer Graphics International</i> (poster), Yokohama, Japan.	2017

Teaching & Mentorship

Mentorship , Modeling and Simulation Lab, GIST: mentored M. Kim from undergraduate intern through graduate studies; mentee published a first-author journal paper (<i>JCDE</i> , 2023). Also guided additional interns and supervised a bachelor’s thesis.	2018–2026
Teaching Assistant , Engineering Analysis, GIST: recitations, assessment design, Python/MATLAB/C++ instruction.	2017, 2020

Professional Service

Reviewer, *Journal of Computational Design and Engineering* (JCDE)

Technical Skills

Languages & Frameworks: Python, PyTorch, C++, MATLAB, Open3D, PCL, Unity3D
Sensors & Devices: Velodyne/Ouster LiDAR, Kinect v2, Azure Kinect, MR smart glasses

References

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